

SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 1.1: Real Numbers — Teacher Reference (Pre-filled)

SUMMARY TABLE: MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 1.1: Real Numbers
Driving Question	DRIVING QUESTION: Why can some numbers be written as exact fractions and others cannot — and how does knowing a number's "type" help us solve everyday problems in Kenya?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 What Kind of Number Is It? — Launching the Pizza Phenomenon	Students observed two pizza diagrams — one divided into 4 equal slices (each = $1/4 = 0.25$, a terminating decimal) and one divided into 3 equal slices (each = $1/3 = 0.3333\dots$, a never-ending repeating decimal). They also encountered $\sqrt{2} \div 2 \approx 0.7071\dots$ as a decimal that never ends and never repeats. Using fraction-circle manipulatives, they recorded decimal equivalents for slices of 2, 3, 4, 5, 6, 8, and 10, noting whether each decimal terminates, repeats, or goes on forever.	Students learned that whole numbers from 1 to 12 can be classified as odd or even (based on divisibility by 2) and as prime or composite (based on the number of factors). They discovered that prime numbers cannot be evenly re-divided into smaller whole-number groups, which connects to why some pizza shares feel 'messy'. They began their Personal Number Classification Map and opened the Driving Question Board with self-generated questions about why some decimals end and others do not.	The TYPE of whole number used to count pizza slices determines whether each person's share can be written as a 'clean' terminating decimal or produces a decimal that goes on forever. Numbers like 2, 4, 5, 8, and 10 produce terminating decimals; 3, 6, and 7 produce repeating or non-terminating decimals. This is connected to whether the denominator has factors of only 2 and 5 — but students have not yet been told this rule formally. The puzzle of WHY this happens, and what it means for 'irrational' numbers like $\sqrt{2}$, is left open as the driving question for the rest of the unit.
Lesson 2 Sorting the Number World: Odd, Even, Prime, and Composite — and the Pizza That Started It All	Students examined two pizza diagrams (one with 4 equal rational slices, one with 3 slices described by a non-terminating decimal) and physically sorted 10 number cards (2, 3, 4, 9, 15, 17, 36, 41, 49, 64) into odd, even, prime, and composite categories using a sorting mat. They also completed a Gallery Walk to compare group sorting decisions.	Whole numbers can be classified as odd (not divisible by 2), even (divisible by 2), prime (exactly two factors: 1 and itself), or composite (more than two factors). The number 1 is neither prime nor composite. The number 2 is uniquely both even and prime. The type of number used to describe pizza slices affects how easily equal sharing can be expressed — prime-slice pizzas are harder to share equally between 2 people without further cutting.	Students connected number classification to the anchoring pizza phenomenon: when a pizza is cut into a prime number of slices (e.g., 3 or 7), equal sharing between 2 people always requires further division, while composite-slice pizzas (e.g., 4 or 8) allow clean equal sharing. Students posted wonder-questions on the DQB and began constructing a cumulative Number World Map that will be revised across all 7 lessons to eventually include rational and irrational

			numbers.
Lesson 3 Reciprocals of Real Numbers — Division, Tables, and Calculators	Learners computed reciprocals of rational numbers ($1/4$, $2/3$, $3/8$, 5 , 0.25) and irrational numbers ($\sqrt{2}$, $\pi/4$) using three methods: long division, mathematical reciprocal tables, and calculators. They recorded and compared results across all three methods in a four-column comparison table. They also observed that the calculator display for irrational reciprocals produces non-terminating, non-repeating decimals.	The reciprocal of any non-zero real number n is $1/n$ — it answers the question 'how many of this piece fit into 1 whole?' The reciprocal of a rational number is always rational (e.g. reciprocal of $2/3$ is $3/2$). The reciprocal of an irrational number is always irrational (e.g. reciprocal of $\sqrt{2}$ is $1/\sqrt{2} = \sqrt{2}/2 \approx 0.7071$). All three methods — division, tables, and calculators — yield the same value, with calculators providing the greatest precision. Reciprocals have real-world applications, such as the mandazi seller problem in Kisumu.	Because we know whether a number is rational or irrational (Lesson 2), we can now predict in advance whether its reciprocal will be neat (a fraction) or messy (an endless non-repeating decimal). This means that for the Nairobi pizza problem: if the slice size is rational, the number of slices that fill the whole is also a rational number — clean and predictable. If the slice size is irrational, the number of slices that fill the whole is also irrational — it exists, but it cannot be expressed as a neat fraction. Knowing a number's type therefore gives us real predictive power about the answers we will get in everyday calculations.
Lesson 4 Rational or Irrational? Placing Real Numbers on the Number Line	Students observed that the decimal expansions of $1 \div 3$ and $3 \div 4$ either terminate or repeat in a predictable pattern, while $\sqrt{2}$ and π , when displayed on a calculator to 10 decimal places, show no repeating block and no termination. Students also physically placed these numbers on a 1-metre number-line strip, observing that $\sqrt{2}$ sits between 1 and 2 (closer to 1.4) and π sits between 3 and 4 (closer to 3.14), visually confirming their magnitude even when their exact fractional form is unknowable.	Students learned that real numbers are classified as rational (expressible as p/q , integers, $q \neq 0$; decimal terminates or repeats) or irrational (cannot be expressed as p/q ; decimal never terminates and never repeats). Key examples: $1/3 = 0.333\dots$ (rational, repeating), 0.75 (rational, terminating), $\sqrt{2} \approx 1.41421356\dots$ (irrational), $\pi \approx 3.14159265\dots$ (irrational). Students also learned that $22/7$, while a common approximation for π , is itself a rational number and is NOT equal to π .	Students explained the pizza-sharing phenomenon using their new classification: pizza slices described by fractions like $1/4$ or $1/8$ are rational — a Nairobi baker can label them exactly. Slices whose size is linked to a measurement involving $\sqrt{2}$ (e.g. the diagonal of a square with side equal to the radius) or to π (arc lengths) are irrational — they exist physically on the pizza but cannot be written as a perfect fraction. The driving question advances: knowing a number's type tells us whether its pizza-slice description is exact or only an approximation.
Lesson 5 Flipping Numbers: Introducing Reciprocals by Division	Students used fraction circle manipulatives to physically show how many equal slices fill a whole pizza, then recorded the relationship in a division table for slice sizes $1/3$, $1/4$, $2/5$, and $3/7$. They also solved the Amina mandazi scenario ($1 \text{ kg flour} \div 2/3 \text{ kg per batch}$) using reciprocal division.	The reciprocal of a number n is $1 \div n$ (or b/a for a fraction a/b), and it represents how many of that quantity fit into 1 whole. The reciprocal exists for all real numbers — including irrationals like $\sqrt{2}$ (reciprocal = $\sqrt{2}/2$) — except zero, which has no reciprocal because division by zero is undefined.	Knowing a number's reciprocal lets us solve everyday division problems in Kenya — such as sharing food equally, adjusting recipes, or measuring quantities in markets — by turning division into multiplication. This works whether the number is rational or irrational, confirming that all real numbers (except zero) behave consistently under this operation, which helps answer why knowing a number's 'type' matters in daily life.
Lesson 6 Applying Reciprocals: Solving Real-World	Students observed that computing the reciprocal of a rational number such as 1 divided by 7 gives consistent results across	Students learned that all three reciprocal methods are valid but serve different purposes: manual division is always	Students explained that the choice of reciprocal method depends on three factors — the type of number (rational or irrational),

Computation Tasks	all three methods, while computing the reciprocal of an irrational number such as 1 divided by root 2 gives slightly different approximations depending on the method used. They also observed that root 4, though written with a radical sign, equals 2 and is rational — its reciprocal is exact.	available and exact for rational numbers; mathematical tables are reliable and portable for up to 4-digit rational values; calculators provide fast approximations including for irrational numbers but always round the result. They learned that irrational reciprocals cannot be expressed exactly by any method.	the tools available (calculator, tables, or neither), and the precision required for the real-world task. They connected this back to the pizza phenomenon by explaining that a slice whose size is irrational — like an arc length involving root 2 — can never be expressed as an exact fraction or terminating decimal, only approximated, just as its reciprocal can only be approximated.
Lesson 7 Final Explanation: Putting It All Together — Real Numbers, Reciprocals, and Everyday Life in Kenya	Learners observed (across the full unit, reviewed in this lesson): two pizza diagrams — one divided into rational-fraction slices ($\frac{1}{4}$, $\frac{1}{3}$, $\frac{1}{8}$) and one where a slice is expressed as an irrational arc length ($\approx 0.7071\dots$). They observed fraction circles as manipulatives showing that some divisions produce terminating decimals and others do not. They observed mathematical table outputs for reciprocals of both rational and irrational numbers, and calculator displays showing non-terminating, non-repeating decimals for $\sqrt{2}$, $\sqrt{3}$, and π .	Learners learned that: real numbers are classified as rational (expressible as $\frac{p}{q}$, where $q \neq 0$) or irrational (non-terminating, non-repeating decimals such as $\sqrt{2}$, $\sqrt{3}$, π); the reciprocal of a number n is $\frac{1}{n}$, found by division, mathematical tables, or calculator; reciprocals are used to solve division problems (e.g., how many slices fit into a whole); knowing a number's type (rational vs. irrational) predicts whether its reciprocal can be written as a neat fraction; whole numbers are further classified as odd, even, prime, and composite (underpinning the rational/irrational distinction).	Learners explained that some pizza slice sizes can be written as exact fractions (rational numbers) because they result from dividing a whole into a countable, finite number of equal parts expressible as $\frac{p}{q}$. Other slice sizes — such as those involving π or $\sqrt{2}$ — are irrational: their decimal expansions never terminate and never repeat, so no exact fraction can capture them. The reciprocal tells us how many of one quantity fit into another (e.g., how many slices fill the whole pizza), and this works for both rational and irrational numbers — but the reciprocal of an irrational number is itself irrational. Knowing a number's type helps Kenyans solve real problems: sharing land equally in the Rift Valley, calculating boda-boda fares per shilling in Kisumu, or dividing githeri portions fairly — because it tells you whether an exact fractional answer exists or whether an approximation is the best you can do.

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